

Reviewer Pack — Minimal Local Indeterminacy in Category Theory and Communica...

Entry 87c2082a2b8e · INCONCLUSIVE

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Minimal Local Indeterminacy in Category Theory and Communication Complexity Rank

Entry ID: 87c2082a2b8e **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-03 14:57:57 UTC

1. Conjecture

Field A (mathematical branch): Category Theory (Monoidal Categories) **Field B** (complexity object): Communication Complexity

Statement:

For any given monoidal category C with n objects, the minimal local indeterminacy of C is linearly correlated with its communication complexity rank $k(C)$, such that $\text{LocalIndet}(C) = \Theta(k(C))$.

Rationale (proposer's reasoning):

The minimal local indeterminacy of a monoidal category could capture structural properties of the category that are related to information flow in communication protocols, potentially providing insights into lower bounds for communication complexity.

Taxonomy category: CATEGORY_THEORY_COMMUNICATION_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 37eedd1fc07bc86d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For any monoidal category C with n objects, $\text{LocalIndet}(C) = \Theta(k(C))$, where $\text{LocalIndet}(C)$ is measured by the Pearson correlation coefficient between local indeterminacy and communication complexity rank, $k(C)$, calculated across multiple random categories.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "monoidal categories" AND "communication complexity" AND local indeterminacy" - "Category Theory" AND monoidal categories AND communication complexity rank" - "minimal local indeterminacy" IN "Category Theory" AND communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_monoidal_category(n):
        # Simple random monoidal category generation for demonstration purposes
        objects = list(range(n))
        morphisms = {}
        for i in range(n):
            for j in range(n):
                morphisms[(i, j)] = [random.randint(0, 1) for _ in range(random.randint(1, 3))]
        return objects, morphisms

    def calculate_local_indeterminacy(morphisms):
        # Placeholder for local indeterminacy calculation
        return sum(sum(morphisms[i][j]) for i in range(len(morphisms)) for j in range(len(morphisms)))

    def calculate_communication_complexity_rank(morphisms):
        # Placeholder for communication complexity rank calculation
        return len(morphisms)

    n = random.randint(5, 40)
    objects, morphisms = generate_monoidal_category(n)

    local_indet = calculate_local_indeterminacy(morphisms)
    comm_complexity_rank = calculate_communication_complexity_rank(morphisms)

    metric_name = "local_indet_vs_comm_complexity"
    metric_value = local_indet / comm_complexity_rank if comm_complexity_rank != 0 else float('inf')
    instances_tested = 1
    n_max = n
    conjecture_holds = False
```

```

counterexample = ""

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys

    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    metric_values = [r["metric_value"] for r in results if not math.isinf(r["metric_value"])]
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(math.isinf(v) for v in metric_values):
        print("RESULT: INCONCLUSIVE reason=metric_saturation")
    elif support_fraction >= 0.8:
        mean_value = sum(metric_values) / len(metric_values)
        std_value = math.sqrt(sum((v - mean_value) ** 2 for v in metric_values) / len(metric_values))
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82d3978c.py", line 67, in <module>
    trial_result = run_trial(seed)
    ~~~~~^~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82d3978c.py", line 41, in run_trial
    local_indet = calculate_local_indeterminacy(morphisms)
    ~~~~~^~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82d3978c.py", line 32, in calculate_local_indeterminacy
    return sum(sum(morphisms[i][j]) for i in range(len(morphisms)) for j in range(len(morphisms)))
    ~~~~~^~~~~~

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82d3978c.py", line 32, in <genexpr>
    return sum(sum(morphisms[i][j]) for i in range(len(morphisms)) for j in range(len(morphisms)))
    ~~~~~^~~~~~

```

KeyError: 0

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating the Pearson correlation coefficient and p-value required to support or falsify the conjecture. | next: Investigate the cause of the crash in the test code and run the test again to obtain the necessary data for a proper evaluation.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	28803
2	preregistration	ollama_remote	glm4:latest	0	0	11772
3	novelty	ollama_remote	glm4:latest	0	0	11463
4	novelty	ollama_remote	glm4:latest	0	0	11759
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14269
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	92569
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22997
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12926
9	judge	ollama_remote	glm4:latest	0	0	12779

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 219338 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/87c2082a2b8e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/87c2082a2b8e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/87c2082a2b8e.tar.gz` (if generated)